

RAKSHIT KUMAR

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Education

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science | CGPA: 9.17

Aug 2023 – May 2027

Chennai, India

Experience

Indian Institute of Information Technology Allahabad

Research and Development Intern

May 2025 – July 2025

On-site

- Developed an **edge-AI system** on **Raspberry Pi 5** with **AI HAT+ (13 TOPS)** for task driven object detection using **TFLite SSD-MobileNet** and **YOLOv11** models.
- Integrated **LLMs (Phi-2, Gemma, Qwen)** via **llama.cpp** with **Chain-of-Thought (CoT)** prompting to reason about object suitability, **increasing model accuracy from 52% to 84%**.
- Built a **real-time PiCamera2 (OpenCV) + Vosk speech system** for multimodal interaction and filtered bounding boxes using LLM responses.
- Benchmarked LLMs across **6 metrics**, monitored system performance with **nmon** and **Matplotlib**, and **plotted comparative graphs**, observed that **Phi-2 reduced inference latency by ~20 seconds compared to Gemma and Qwen** on Raspberry Pi 5.

Elevance Skills

Full Stack Web Development Intern

Dec 2025 – Feb 2026

Remote

- Developed a full-stack web application using **Next.js, React, TypeScript, Node.js, Express.js, and MongoDB**, implementing **secure authentication, role-based access control, and real-world business workflows**.
- Built enterprise-style security and user management features including **OTP-based login, Google OAuth, Firebase authentication, device tracking, login history monitoring, and browser-based access control**.
- Designed scalable backend APIs and integrated advanced functionalities such as **Multilingual UI support, Multimedia uploads, Browser notifications, Subscription plans, and payment/media services** using **Razorpay and Clouinary**.

Projects

Integrated Flood Prediction & Smart Evacuation System

Nov 2025 – March 2026

- Developed an end-to-end flood prediction and GIS-based evacuation routing system using **XGBoost, LSTM, Python, and QGIS**, trained on **10+ years of rainfall, runoff, and soil moisture data**, with **Chennai as the primary study region**.
- Generated flood risk zonation using the **HAND model** and dynamically filtered safe shelters for disaster response planning.
- Implemented network-based shortest path optimization to compute optimal evacuation routes during flood scenarios.

Stock Market Price Predictor

July 2025 – Sep 2025

- Developed a **ML/DL** forecasting pipeline (Random Forest, SVR, Lasso/Ridge, Gradient Boosting, MLP, LSTM, GRU) on Adani Ports dataset to generate short- & long-term stock price predictions, evaluated via MSE, MAE & R².
- Integrated **sentiment analysis** from ~1.39M financial tweets with technical indicators to enhance model robustness and capture market psychology.
- Built end-to-end workflows (data cleaning, feature engineering, scaling, visualization, benchmarking) to streamline model comparison and deliver investor-ready insights.

Real-Time Multiplayer Chess Game

June 2025

- Developed a web-based multiplayer chess game enabling real-time gameplay using **Socket.io** for bidirectional WebSocket communication and **achieved 100% move validation accuracy, game logic using Chess.js**.
- Designed a role-based player system (white, black, spectator) with synchronized game state across clients using **FEN notation** and validated across **50+ users**.
- Built an interactive drag-and-drop chessboard interface with **Node.js**, styled with **Tailwind CSS**, and rendered using **EJS templating and Express.js**.

Skills

Languages: HTML, CSS, Tailwind CSS, JavaScript, C, C++, Python, TypeScript

Frameworks/Libraries: React.js, Node.js, Express.js, Redux, OpenCV, NumPy, Next.js

Technical Skills: Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Machine Learning

Tools/Databases: Git, GitHub, MySQL, MongoDB, Firebase, Clouinary

Soft Skills: Analytical Thinking, Problem Solving, Communication & Presentation, Leadership, Collaboration, Adaptability, Business Acumen, Entrepreneurship

Achievements and Leadership

- Award of Excellence during IIT Allahabad internship**, honored for outstanding contribution in building an edge-AI task driven object detection system with Raspberry Pi 5, AI HAT+ (13 TOPS) and LLM integration.
- Founder, DRIFTZ Clothing Brand (Business)** - Built and managed DRIFTZ, a premium streetwear clothing brand, achieving **sales of 3 lakhs+ rupees** and successfully **selling 400+ oversized T-shirts**. Oversaw product design, branding, marketing, customer targeting, and business operations from concept to execution.
- Technical Head, Liftoff Club** - Led web design and development initiatives for the club, enhancing digital engagement while mentoring developers through code reviews and technical guidance. Collaborated with cross-functional clubs to organize workshops, technical events, and learning sessions, fostering skill development across members.